

Varshitha Thilak Kumar

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[My Portfolio](#)

[Varshitha Thilak Kumar | LinkedIn](#)

[Varshitha Thilak Kumar | GitHub](#)



Skills

- Computational Thinking & Problem Solving
- ML and basics of DL
- C and C++ Programming
- Data Structures & Algorithms (DSA)
- MATLAB(Basics)
- Python Programming
- Web Development
- Languages Known:English,Tamil,Telugu,Hindi

Projects

1. Automated Battery Materials Data Extraction from Scientific Literature using LLMs and Ex AI

- Developed an automated pipeline leveraging **LLMs, PDF processing, and SHAP-based Explainable AI** to extract and normalize structured battery property data (capacity, voltage, conductivity) from scientific literature with 90–95% accuracy.
- Faced challenges in **accurately parsing unstructured scientific PDFs and aligning diverse measurement units**, resolved through hybrid **pattern matching + LLM extraction** and standardized data normalization.

2. DDoS detection in SDN using ML

- Developed a DDoS detection and mitigation system in SDN using Mininet, integrating ML models (Random Forest, SVM) with a Ryu controller for real-time traffic classification.
- Implemented automated blocking and rerouting strategies to enhance network resilience against evolving threats.

3. Movie Inferno – Movie Recommendation Platform (DBMS + Cloud)

- Built a **Movie Management System** using **Next.js, Supabase, and TMDB API**, integrated with an **AI-powered chatbot** for personalized movie recommendations, and deployed on **Vercel** with real-time updates.
- Initial **deployment errors on Vercel** due to environment variable misconfiguration; debugged build logs and securely managed API keys through **.env** files.

4. Time-Frequency Analysis on PQD using Advanced Signal Processing Techniques (MATLAB)

- Applied advanced signal processing techniques for power quality disturbance analysis.
- Faced difficulty in understanding the mathematical formulas for signal processing methods.

5. Phishing Website Detection (Machine Learning)

- Developed a phishing website detection system using machine learning models.
- Analyzed a dataset containing legitimate and phishing websites.
- Improved feature extraction techniques to enhance classification accuracy.

Education

Maharishi Vidya Mandir / 10th Graduation

June 2020 - March 2021, Chennai

Passed out with **93.6%**

Velammal Vidyalaya / 12th Graduation

June 2021 - March 2023, Chennai

Passed out with **87.2%**

Amrita Vishwa Vidyapeetham / B-Tech CSE-AI

August 2023 - July 2027, Coimbatore

Currently pursuing 3rd Year

CGPA: 8.5

Awards and Certifications

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- Presented Paper at IEEE Conference (ETIS, Trivandrum - 2024)
Topic: Time-Frequency Analysis on PQD using Advanced Signal Processing Techniques (MATLAB)
Published on IEEE Xplore-<https://ieeexplore.ieee.org/document/10961578>
 - **Finalist, Cubicle Code 5.0 Hackathon, 2025** — Conducted by Geek Rooms in collaboration with Pathway and TRAE at Microsoft, Bangalore
 - **Finalist in GenAI Hackathon (Value Health Org) - 2024**
Developed GenJournals, an AI-powered journaling application.
 - Completed Full Stack Development course (MERN stack) at IITM research park through Copter Code
 - Completed Gen AI Course(Vertex AI, Gemini APIs, Imagen, Streamlit and Multimodal RAG) by Gen AI Academy that helped be gain foundational knowledge and skills using Google's Gen AI tools and Platforms
 - Oracle Cloud Infrastructure 2025 Certified Foundation Associate
 - Certified in C and C++ (FIIT)